

An eigenvalue-weighted extension of branch saturation testing

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Abstract

Branch saturation tests ask whether two subtrees connected by a branch still share phylogenetic signal. SatuTe answers this question with a coherence coefficient for the dominant non-stationary eigenvalue of a reversible substitution model. This draft defines an eigenvalue-weighted extension that keeps the same subtree likelihood notation and null test, but replaces a single coefficient by a branch-length-aware sum over non-stationary coherence coefficients. The extension is identical to SatuTe under Jukes–Cantor symmetry and gives a paired general time-reversible comparison in which eigenvectors describe directions of signal and eigenvalues determine the expected persistence of those directions.

Saturation is a practical failure mode for phylogenomic inference. A branch can be present in a maximum-likelihood tree, and can even have high conventional support, while the alignment no longer contains information that reliably connects the two subtrees separated by that branch. SatuTe formalized this problem as a branch-specific test of subtree independence: for a branch AB , the subtrees T_A and T_B are treated as phylogenetically informative if their subpatterns ∂_A and ∂_B show shared signal beyond what is expected under independent evolution [1]. This reframes saturation from a pairwise sequence property into a testable property of a split in a tree.

The existing statistic uses the dominant non-stationary eigenspace. For a site pattern ∂ , SatuTe computes the partial likelihood vectors $L(\partial_A)$ and $L(\partial_B)$, the subtree pattern probabilities $P(\partial_A) = \langle L(\partial_A), \pi \rangle$ and $P(\partial_B) = \langle L(\partial_B), \pi \rangle$, and the coherence coefficient associated with the second largest eigenvalue of the rate matrix. With eigenvalues $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \lambda_3$ and left eigenvectors h_i , the published coefficient for a simple eigenvalue is

$$C_{\partial i} = \left\langle \frac{L(\partial_A)}{P(\partial_A)}, h_i \right\rangle \left\langle \frac{L(\partial_B)}{P(\partial_B)}, h_i \right\rangle. \quad (1)$$

For the dominant non-stationary mode, the empirical statistic is $\hat{C}_1 = n^{-1} \sum_{s=1}^n C_{\partial_s 1}$, with variance estimator $\hat{\sigma}_1^2$ and one-sided statistic $Z = \sqrt{n} \hat{C}_1 / \hat{\sigma}_1$. Rejecting $Z \leq z_\alpha$ is not the aim; SatuTe rejects subtree independence when $Z > z_\alpha$, and therefore calls branch AB phylogenetically informative. Failure to reject means that AB is treated as saturated.

This construction tests whether the dominant surviving spectral direction links T_A and T_B . Under a general time-reversible (GTR) model, λ_1 , λ_2 and λ_3 are usually distinct. Their eigenvectors h_i describe different directions in nucleotide state space, while their eigenvalues describe different decay rates along branch AB . A statistic that uses only $C_{\partial 1}$ can omit signal carried by h_2 or h_3 . The weighted statistic tested here allows non-dominant directions to contribute only through the relative decay implied by the fitted model.

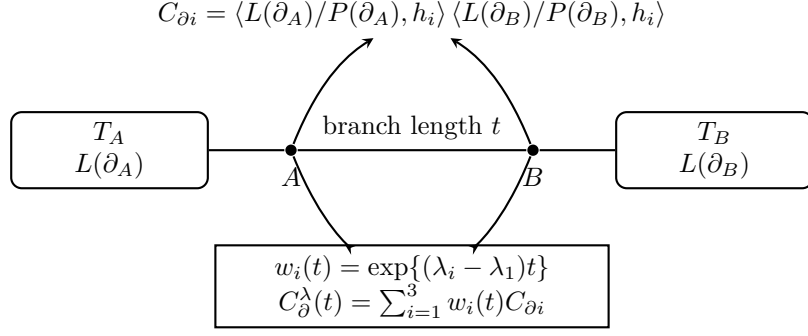


Figure 1: **Eigenvalue weighting in SatuTe notation.** Branch AB splits a site pattern ∂ into ∂_A and ∂_B . SatuTe computes coherence coefficients $C_{\partial i}$ from normalized partial likelihood vectors and eigenvectors h_i . The extension keeps these coefficients but weights each non-stationary mode by its relative persistence over branch length t .

Results

A weighted coefficient extends the SatuTe statistic

The spectral decomposition used by SatuTe gives

$$\frac{P(\partial | t)}{P(\partial_A)P(\partial_B)} = 1 + \sum_{i=1}^3 C_{\partial i} \exp(\lambda_i t). \quad (2)$$

Equation (2) contains both the coherence coefficients and their branch-length decay terms. The coefficient $C_{\partial i}$ measures whether the two subtrees agree in the spectral direction h_i . The term $\exp(\lambda_i t)$ measures how much of that direction remains after evolution across branch AB . Because all non-stationary eigenvalues are negative, the modes with more negative λ_i decay faster.

We define the eigenvalue-weighted coherence for pattern ∂ as

$$C_{\partial}^{\lambda}(t) = \sum_{i=1}^3 w_i(t) C_{\partial i}, \quad w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}. \quad (3)$$

The relative weight $w_i(t)$ sets $w_1(t) = 1$ and downweights modes that decay faster than the dominant non-stationary mode. This normalization keeps the published SatuTe coefficient as the reference scale while allowing additional eigenvectors to contribute when their expected persistence across branch AB is non-negligible.

For an alignment of n independently evolving sites, the empirical coefficient is

$$\hat{C}_{\lambda}(t) = \frac{1}{n} \sum_{s=1}^n C_{\partial_s}^{\lambda}(t). \quad (4)$$

The variance estimator follows the same structure as the SatuTe estimator for a repeated second largest eigenvalue, but includes the mode weights:

$$\hat{\sigma}_{\lambda}^2(t) = \sum_{i=1}^3 \sum_{j=1}^3 w_i(t) w_j(t) \hat{\sigma}_{A,i,j}^2 \hat{\sigma}_{B,i,j}^2. \quad (5)$$

The corresponding test statistic is

$$Z_{\lambda}(t) = \frac{\sqrt{n} \hat{C}_{\lambda}(t)}{\hat{\sigma}_{\lambda}(t)}. \quad (6)$$

The decision rule is unchanged: branch AB is called phylogenetically informative when $Z_{\lambda}(t) > z_{\alpha}$, or when $Z_{\lambda}(t) > z_{\alpha/(m_A m_B)}$ under the same Bonferroni correction used for maximum-likelihood tree double dipping.

The Jukes–Cantor case is unchanged

The Jukes–Cantor (JC) model has one stationary eigenvalue and a three-dimensional non-stationary eigenspace with $\lambda_1 = \lambda_2 = \lambda_3$. Therefore $w_i(t) = 1$ for all non-stationary modes in Eq. (3). The weighted statistic becomes

$$C_{\partial}^{\lambda}(t) = \sum_{k=1}^3 C_{\partial 1, k} = C_{\partial 1}, \quad (7)$$

which is the multiplicity-three expression used by SatuTe. Thus, the weighted statistic does not change the JC statistic. When the substitution model does not distinguish among the non-stationary directions, eigenvalue weighting does not introduce a distinction.

GTR separates direction from persistence

Under a typical GTR model, the second largest eigenvalue has multiplicity one. The published statistic therefore uses $C_{\partial 1}$, the most persistent non-stationary direction. This choice follows the slowest approach to stationarity. It can, however, discard signal in h_2 and h_3 when branch AB is short enough that those directions remain visible.

The weighted statistic in Eq. (3) separates direction from persistence. For short branches, $w_i(t)$ remains close to one and multiple directions can contribute. For long branches, $w_i(t)$ approaches zero for faster modes and the statistic approaches the dominant SatuTe coefficient.

The original SatuTe simulations define the baseline

The manuscript archive fixes the baseline simulation design more precisely than the article text alone. For the five-taxon simulation, the tested external branch AB connects taxon B to the root A of a balanced four-taxon subtree T_A . All branches inside T_A have length 0.2, and the length of AB is varied. For the 16-taxon simulation, AB connects the roots A and B of two balanced eight-taxon subtrees. Branch lengths inside these subtrees are not fixed; external and internal branches are sampled from EvoNAPS-derived empirical branch-length distributions in the interval 0.04 to 0.2. In both cases, the branch length of AB is varied over the same 16-value grid, and 1,000 DNA alignments are generated for each combination of tree case, alignment length and branch length under the JC model with Seq-Gen.

The four analysis scenarios are also part of the statistical design, not merely plotting choices. Each simulated alignment is analyzed on the true tree with fixed true branch lengths, on the true topology with maximum-likelihood branch lengths, on a maximum-likelihood inferred tree, and on the same maximum-likelihood tree with a Bonferroni-corrected threshold. The correction is $\alpha/4$ for the five-taxon external branch and $\alpha/64$ for the 16-taxon internal branch. In the original 16-taxon analysis, maximum-likelihood trees that failed to recover the split induced by AB were discarded before the fraction of informative instances was computed. This discard rule must be preserved in any direct comparison to the published curves.

Validation must match the original simulation design

The empirical comparison cannot be based on a small reduced run. A reduced run can test software wiring, branch selection and file parsing, but it cannot support claims about power, calibration or trustworthiness. The validation for this extension therefore starts by reproducing the SatuTe simulation design before adding GTR alternatives. In the JC baseline, branch length AB varies over $\{0.1, 0.2, 0.3, 0.4, 0.5, 0.8, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 5.0, 7.5, 10.0\}$, with 1,000 alignments per parameter combination and sequence lengths $n = 100, 1,000, 10,000$. The five-taxon case uses the $B \mid A_1, A_2, A_3, A_4$ split described in the manuscript archive. The 16-taxon

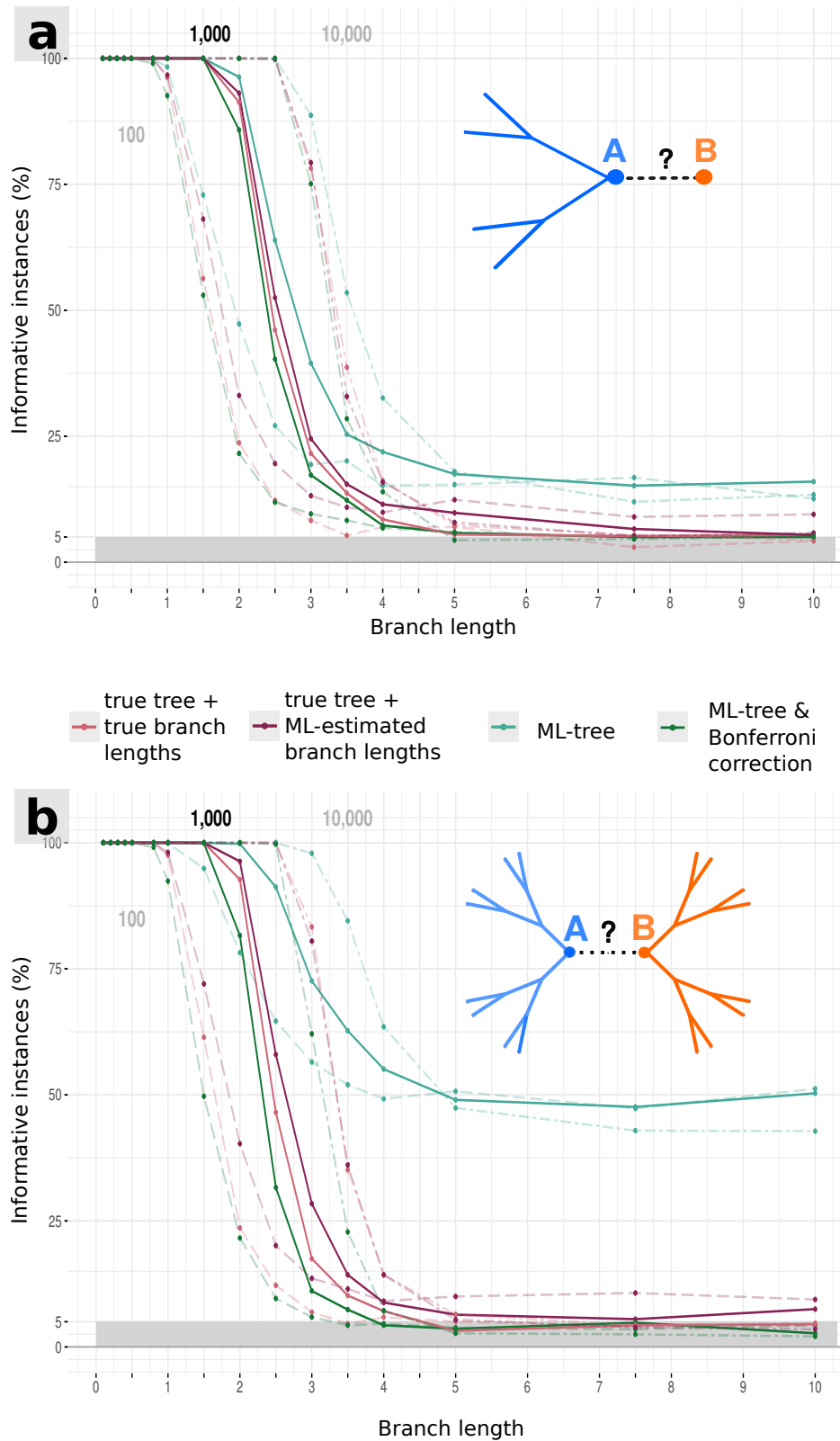


Figure 2: **Original SatuTe simulation figure used as the reproduction target.** The extension analysis uses the same plotting structure: two tree cases, four analysis scenarios, three sequence lengths and the same branch-length grid. The figure is copied from the manuscript archive and is not regenerated from the local development output.

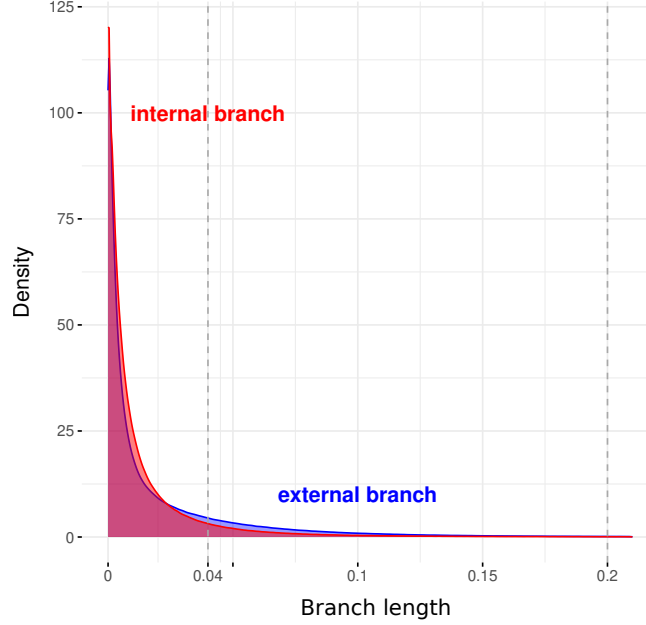


Figure 3: **EvoNAPS branch-length distribution required for the 16-taxon simulations.** A direct rerun samples internal and external subtree branches from the empirical pools represented here. Fixed 0.1 subtree branches are used only for software development checks.

case uses the central $A_1, \dots, A_8 \mid B_1, \dots, B_8$ split with subtree branch lengths sampled from the EvoNAPS empirical distributions.

After the JC baseline reproduces the published behavior, the extension can be evaluated under GTR. The comparison is paired. One simulated alignment, one random seed, one analysis scenario and one target branch define a replicate. On that same replicate, the analysis computes the published dominant coefficient $C_{\partial 1}$ and the eigenvalue-weighted coefficient $C_{\partial}^{\lambda}(t)$. The same one-sided normal test, the same α , and the same $m_A m_B$ Bonferroni threshold are then applied to both statistics. The endpoints are type I error under subtree independence, power under known informative branches, behavior under maximum-likelihood branch-length estimation, and inflation under maximum-likelihood topology selection with and without the Bonferroni correction.

The paired design changes the reporting. Power curves still show the fraction of replicates called informative, but each parameter combination should also include a discordance table: dominant-only, weighted-only and agreement. McNemar tests on paired decisions and paired summaries of $Z_{\lambda}(t) - Z_1$ or $-\log_{10} P_{\lambda} + \log_{10} P_1$ compare formulas on the same simulated alignments and remove simulation noise shared by both formulas.

Biological sliding-window rerun

The curated biological rerun applies the same extension to the manuscript 16S rRNA Tree-of-Life dataset. The dataset is the rRNA-based 3D tree with 1,947 alignment sites, a fitted GTR+F+G4 model, four rate categories and 36-site sliding windows. The focal branches are the internal branch leading to Eukaryota and the external branch leading to yeast. The comparison keeps the published SatuTe rerun, the native dominant-mode calculation and the eigenvalue-weighted statistic.

For the branch leading to Eukaryota, the published SatuTe rerun marks 822 of 1,912 windows as saturated. The native dominant calculation marks 895 windows, whereas the eigenvalue-weighted statistic marks 699 windows. For the yeast branch, the corresponding counts are 70, 83

Table 1: **Validation grid required for empirical claims.** The reduced files in this repository are development checks. They are not simulation evidence for power or calibration.

Component	JC baseline	GTR enhancement
Tree cases	Five-taxon external; 16-taxon internal	Same tree cases
Branch lengths	Published 16-value grid	Same grid
Sequence lengths	100, 1,000, 10,000	Same lengths
Replicates	1,000 per combination	1,000 per combination
Simulator	Seq-Gen-compatible JC design	GTR matrices with controlled spectra
Scenarios	True fixed, true ML lengths, ML tree, Bonferroni	Same scenarios
Replicate unit	One alignment, one scenario, one target branch	Same replicate for both formulas
Test threshold	Same α , same Bonferroni rule	Same α , same Bonferroni rule
Statistics	$C_{\partial 1}$	$C_{\partial 1}, C_{\partial}^{\lambda}(t)$
Main comparison	Calibration of SatuTe baseline	Paired decisions and paired Z-score differences

and 29 saturated windows. The difference is therefore concentrated in the same sliding-window setting used by the original paper, rather than in a simulated example.

A second biological rerun uses an independent implementation that does not read SatuTe component files or SatuTe result trees. It recomputes the sliding-window scores from the IQ-TREE treefile, model report, site-probability output and the original alignments. This run covers both Tree-of-Life examples from the manuscript: the ribosomal protein alignment on the protein-based 2D tree under LG+G4, and the 16S rRNA alignment on the rRNA-based 3D tree under GTR+G4. In the protein 2D tree, the branch leading to Eukaryota has 2,561 windows, with 138 saturated windows in the published SatuTe rerun and 5 in the eigenvalue-weighted calculation. The yeast branch also has 2,561 windows, with counts 54 and 8. In the 16S rRNA 3D tree, the branch leading to Eukaryota has 1,912 windows, with counts 822 and 699. The yeast branch has 1,912 windows, with counts 70 and 29.

Discussion

The extension changes the spectral aggregation step while keeping the SatuTe null test. SatuTe already derives branch saturation from the spectral decomposition of $\text{Diag}(\pi)P(t)$. The weighted statistic asks whether the coefficient tested by SatuTe should remain restricted to the single slowest mode under GTR, or whether other model-defined directions should contribute according to their expected persistence over branch AB . Because the formula is built from $C_{\partial i}$, λ_i , $\hat{\sigma}_{A,i,j}^2$, $\hat{\sigma}_{B,i,j}^2$ and Z , it keeps the notation and null logic of the original method.

This distinction can be framed as a calibration problem. The published $C_{\partial 1}$ statistic is stringent in GTR because it tests only the slowest surviving non-stationary direction. The eigenvalue-weighted statistic allows additional eigenvectors to contribute, but scales each contribution by the decay rate implied by the fitted model. On long branches, this makes the method approach the dominant SatuTe statistic. On short branches, non-dominant directions can still contribute.

The method is an extension rather than a replacement. The dominant SatuTe coefficient remains the reference statistic for branch saturation in IQ-TREE until the weighted statistic has been calibrated under the full simulation grid. If calibration supports the weighted test, it would add a model-aware diagnostic: not only whether two subtrees agree, but whether they agree in spectral directions that the fitted substitution model predicts should remain visible across the

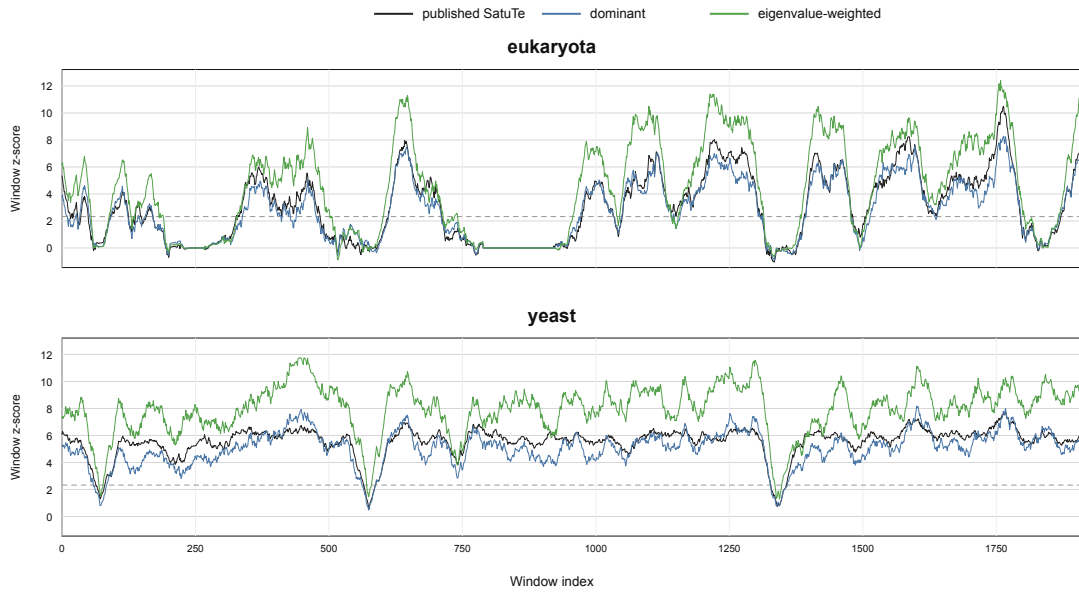


Figure 4: **Published SatuTe and eigenvalue-weighted sliding-window curves for the 16S rRNA Tree-of-Life example.** The curves compare the manuscript SatuTe rerun with the native dominant calculation and the eigenvalue-weighted statistic for the branches leading to Eukaryota and yeast.

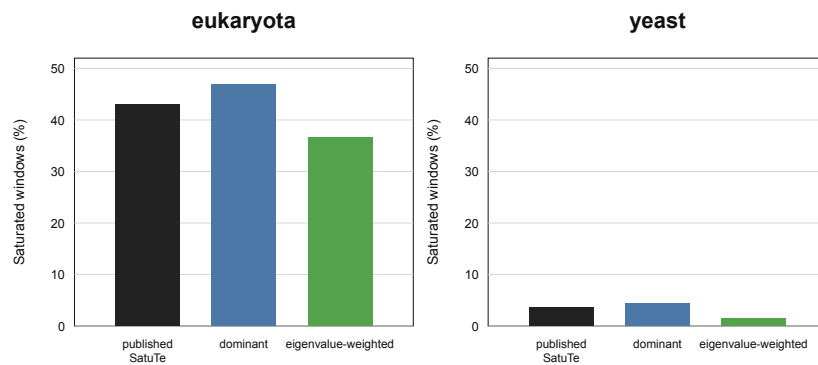


Figure 5: **Number of saturated sliding windows in the 16S rRNA Tree-of-Life example.** The eigenvalue-weighted statistic changes the number of windows classified as saturated while using the same 36-site windows and the same branch targets.

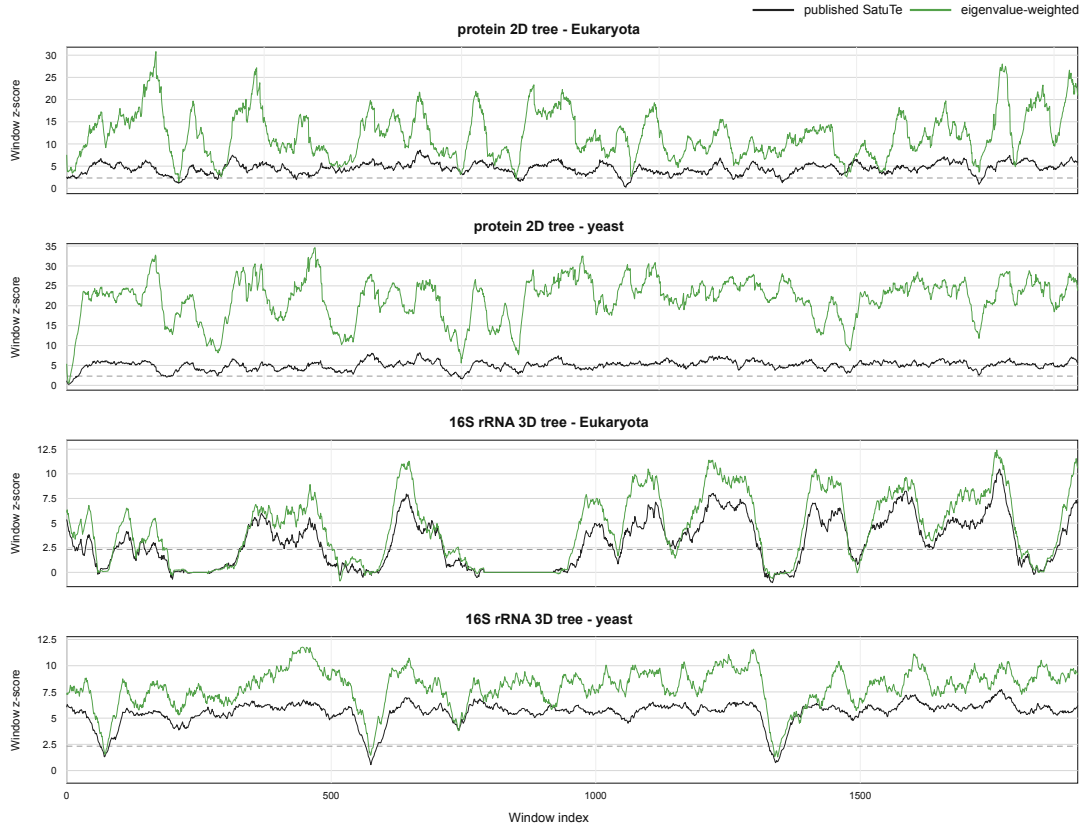


Figure 6: **Independent eigenvalue-weighted sliding-window curves for the protein 2D and 16S rRNA 3D Tree-of-Life examples.** The independent calculation compares the published SatuTe sliding-window score with the eigenvalue-weighted statistic on the same 36-site windows and target branches.

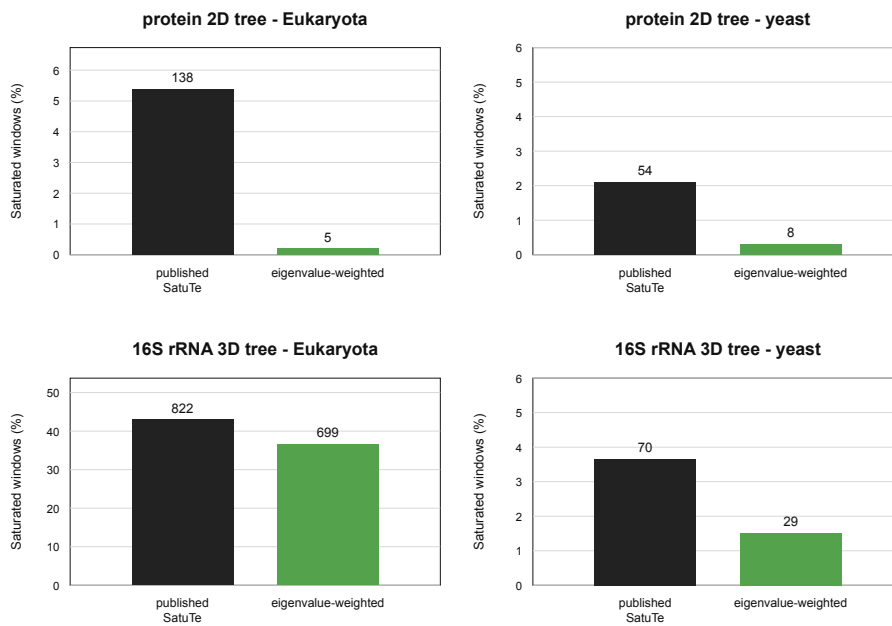


Figure 7: **Saturated-window counts in the independent biological comparison.** Counts are shown for the branch leading to Eukaryota and the branch leading to yeast in the protein-based 2D tree and the 16S rRNA-based 3D tree.

focal branch.

Online Methods

Notation

The notation follows SatuTe. A branch AB separates subtrees T_A and T_B . A site pattern ∂ induces subpatterns ∂_A and ∂_B . The partial likelihood vectors at the endpoints are $L(\partial_A)$ and $L(\partial_B)$. The stationary distribution is π , and $P(\partial_A) = \langle L(\partial_A), \pi \rangle$, $P(\partial_B) = \langle L(\partial_B), \pi \rangle$. The reversible rate matrix has eigenvalues $\lambda_0 = 0 > \lambda_1 \geq \lambda_2 \geq \lambda_3$ and left eigenvectors h_i . Coherence coefficients are defined by Eq. (1).

Weighted estimator

For each site s , compute $C_{\partial_s i}$ for the non-stationary modes. For branch length t , compute $w_i(t) = \exp\{(\lambda_i - \lambda_1)t\}$. Then compute $\hat{C}_\lambda(t)$, $\hat{\sigma}_\lambda^2(t)$ and $Z_\lambda(t)$ using Eqs. (4) to (6). If a model has a repeated second largest eigenvalue, each direction in that eigenspace has weight one. For JC, this gives the published SatuTe statistic.

Simulation protocol

The baseline simulation first reproduces the published JC design. DNA alignments are generated with Seq-Gen, using the same tree cases, branch-length grid, sequence lengths and replicate count used for SatuTe. The five-taxon tree is a balanced four-taxon subtree T_A with all subtree branches set to 0.2 and a single taxon B attached by branch AB . The 16-taxon tree is built from two balanced eight-taxon subtrees whose internal and external branch lengths are sampled from EvoNAPS-derived pools in the interval 0.04 to 0.2. Each alignment is analyzed under four scenarios: the true simulation tree with fixed true branch lengths, the true topology with maximum-likelihood branch lengths, the maximum-likelihood inferred tree, and the maximum-likelihood inferred tree with Bonferroni threshold $\alpha/(m_A m_B)$. The GTR extension then repeats the same topologies and scenario logic with rate matrices chosen to span weak, moderate and strong separation between λ_1 , λ_2 and λ_3 . Within each replicate, both formulas are evaluated on the same inferred or supplied tree, the same estimated branch length and the same target split.

Implementation status

The local rerun script implements the paired design and the paper topologies. It records the simulator, branch-length source, target split, scenario, formula, Z -score, P -value and decision for every replicate. The script supports a direct rerun mode with Seq-Gen and an external EvoNAPS branch-length table, and a development-check mode using IQ-TREE AliSim with fixed 0.1 subtree branches when the raw EvoNAPS branch-length pools are not available. The development-check mode tests the IQ-TREE integration and formula comparison, but it is not evidence for the manuscript claims. The biological sliding-window reruns are stored with the curated biological results accompanying this manuscript.

An exact rerun still requires three external inputs that are not present in the manuscript archive: the Seq-Gen binary, the raw EvoNAPS internal and external branch-length samples used for the 16-taxon simulations, and TreeShredder or an equivalent split-checking procedure to discard maximum-likelihood trees that do not recover the AB split. Without these dependencies, a run can verify code wiring but cannot claim to reproduce the original SatuTe simulation curves.

Data availability

No full-scale simulation data are reported in this draft. The SatuTe source paper, the manuscript archive and public repositories are available from the cited sources. The local repository contains reduced simulation outputs used only to verify command wiring, branch selection and paired formula evaluation, and it contains the curated biological reruns used for the sliding-window figures.

Code availability

The working IQ-TREE integration and manuscript sources are in the local repository. A publication version should archive the exact simulation scripts, IQ-TREE commit, SatuTe command lines, random seeds and generated summary tables.

Acknowledgements

This manuscript draft builds on the SatuTe framework of Manuel and colleagues and on the IQ-TREE integration work in this repository.

Author contributions

B.S. developed the enhancement concept and prepared the manuscript draft.

Competing interests

The author declares no competing interests in this draft.

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